



Grade 7 Mathematics

OPTIMA ACADEMY ONLINE · FAMILY COURSE OVERVIEW · M/J GRADE 7 MATH · FLORIDA COURSE 1205040 · FULL YEAR

~4¼ hrs per week
50 min/day, M–F

About this overview. This document is an example of how the course might be taught. Scholars may see some variation in the lessons, examples, and practice sets from course to course, and between the live school and On-Demand. Each teacher also leans into their own strengths and passions; the structure, expectations, and time commitment below are representative of the course as designed.

Grade 7 Mathematics centers on proportional reasoning, the key idea of middle school math. Scholars sharpen their work with rational numbers and exponents, learn to solve linear equations and inequalities, and master percentages and proportional relationships, including the discounts, markups, interest, and conversions they will meet in daily life. The year closes with statistics, probability, and geometry, applying ratio thinking to data, chance, scale drawings, and circles.

THE YEAR AT A GLANCE

M1 Rational Numbers & Laws of Exponents (~2½ wks)
numbers in different forms; exponent properties

M3 Linear Expressions, Equations & Inequalities (~2 wks)
solve one- and two-step equations

M5 Proportional Relationships (~2½ wks)
constant of proportionality; graphs and equations

M7 Simple Experiments & Probability (~2½ wks)
sample spaces; theoretical vs experimental

M9 Cylinders (~2 wks)
surface area and volume problems

M2 Operations on Rational Numbers (~2 wks)
the four operations; order of operations

M4 Percentages & Proportional Reasoning (~2½ wks)
discounts, markups, interest, conversions

M6 Representing Statistical Data (~4 wks)
center and variation, circle graphs, comparing data

M8 Two-Dimensional Figures & Scaling (~3 wks)
area, circles and pi, scale drawings

A TYPICAL WEEK

- A new lesson most days: teacher-guided instruction with worked examples and notes (about 30–35 min).
- Practice problems with every lesson, worked out on paper, to make each skill automatic (about 15–20 min).
- A review and module assessment as each module closes, roughly every two to three weeks.

TIME COMMITMENT

250 minutes / week

≈ 50 minutes a day, Monday–Friday

- Daily lesson + practice fills most of the 50 minutes.
- Module review and assessment days replace new lessons as each module closes.

WHAT SCHOLARS NEED

- A computer with reliable internet and access to Canvas.
- A dedicated math notebook, pencil, and paper — every problem gets worked by hand.
- A basic calculator is useful for checking work; the goal is fluency by hand first.
- Microsoft 365 and OneDrive (provided to every scholar) for any typed work.

On-Demand (asynchronous). Scholars move through the course at their own pace, with due dates to keep the year on track. An Optima teacher is available throughout for support, feedback, and grading.

Live school. Live-school scholars take this same course on Optima's VR campus with a teacher and classmates, meeting daily as a core course.